

➤ **Title of the Project:**

On the static and dynamic motion of liquid motion on solid surfaces

➤ **Project Descriptions:**

Moving liquid on solid appears ubiquitously in nature and various industrial applications. In recent years, the digital microfluidic technology, by using tiny drops as carrier to enable effective transportation and high-efficient reaction performance, has been widely studied and applied in research and applications. Designs of different surface properties, e.g., geometry, wettability or roughness etc., have been widely used in the advanced applications. However, the existing data about the onset motion of liquid, e.g., a liquid drop, on the solid surfaces are quite diverse (up to a factor of 2) and not comparable for even the same condition of drop and solid substrate. The theoretical description and its mechanism of a drop starting to move on a solid substrate are also still not fully satisfactory and clear, especially for a drop moving across surfaces with different properties. Due to the complexity of the related controlling parameters, a systematic analysis together with a reliable databank is expected to solve the long-existing puzzle and to screen different proposed theories in the literature. In this project, we'll first develop a novel measurement platform, consisting of the high-precision and high reliability for wide range of drop-size tests. The static/dynamic contact parameters and the related drop morphologies will be measured and analyzed. The study will be mainly investigated and analyzed experimentally in the project. The tools of simulation and theoretical modelling can be also conducted in parallel depending on the progress of the study and the background of the intern.

In this project, the intern can learn the manufacturing techniques of surface treatment of superhydrophobic substrates and the skill of precision instrumentation

design and control technologies within a period of 2-3 months. Furthermore, the intern can learn how to analyze the measured data to understand the liquid moving mechanism and how to prepare the good presentation and systematical report. All candidates with enthusiastic heart for the scientific research are warmly welcome. Those who, majoring in physics and engineering or having experience in experiment, are especially encouraged. The research schedule of intern will be planned as follows,

- (I) Basic lab training and practice of machining, such as laser engraving, CNC machining etc., for the fabrication of the needed experimental setup. (1-2 weeks)
- (II) Training of fluid properties measurement, high-speed camera & visualization practice and data-acquisition & processing. (1-2 weeks)
- (III) Surface treatment training, e.g., different wettability, contact angle derivation, visualization and digitization of drop motion on surface with different controlling parameters, e.g., plane inclined angles, wetting conditions, moving speeds and fluid properties experiments. (4-6 weeks)
- (IV) Motion control of the digital microfluidics and its application design (2-3 weeks)
- (V) Mechanical mechanism and results discussion, and report finalization (1 week)

➤ **Responsibilities:**

- 1. Obey the rules of lab and instructions of senior members.
- 2. Join the assigned training program, meeting and finish a complete internship report.

➤ **Requirements:**

1. Good teamwork and English communication ability (Communication in German is only possible to the lab director, Prof. Wang, in the lab).
2. Basic knowledge of physics and experimental ability.
3. Enthusiasm for scientific research.

➤ **Information**

1. Lab: <http://bernoulli.iam.ntu.edu.tw/>
2. Institute of Applied Mechanics (IAM): <http://www.iam.ntu.edu.tw/en/>
3. National Taiwan University (NTU): <http://www.ntu.edu.tw/english/>
(We will help for finding the accommodation of qualified interns.)
4. Taipei tourist information:
<https://eng.taiwan.net.tw/m1.aspx?sNo=0002090>

Keywords: Digital microfluidics, Drop statics & dynamics, Liquid-solid interface friction, Surface properties, Measuring technique, Moving mechanism